Question	Answer	Marks	Guidance
1(a)	(electric) force is (directly) proportional to product of charges	B1	**Ignore** any symbols unless they are defined.
	(electric) force (between point charges) is inversely proportional to the square of their separation	B1	**Ignore** any symbols unless they are defined. **Do not allow** just 'distance' for 'separation' unless it is clear that the distance is between the charges. **Do not allow** 'radius' for 'separation'.
1(b)(i)	$(-)\ (1.09 \times 10^{-18}) = (-)\ (1.60 \times 10^{-19})^2 /$ $(4\pi \times 8.85 \times 10^{-12} \times R)$ leading to $R = 2.11 \times 10^{-10}$ m	A1	**Allow** '8.99×10^{9}' for '$1 / (4\pi \times 8.85 \times 10^{-12})$' Full substitution and answer needed. **Ignore** signs of the energies. Unit not needed but any unit given must be correct.
1(b)(ii)	$F = (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2.11 \times 10^{-10})^2]$ $= 5.17 \times 10^{-9}$ N	A1	No ECF from **1(b)(i)**. **Allow** 8.99×10^{9} for $1 / (4\pi \times 8.85 \times 10^{-12})$ Correct to at least 3 significant figures. AFC. **Allow** 5.16×10^{-9} N AFC **Ignore** the sign of the answer.
1(b)(iii)	$a = (5.17 \times 10^{-9}) / (9.11 \times 10^{-31})$ $= 5.68 \times 10^{21}$ m s^{-2}	A1	Possible ECF from **1(b)(ii)**. Correct to at least 3 significant figures. AFC. **Allow** 5.67×10^{21} m s^{-2} or 5.66×10^{21} m s^{-2} as AFC answers
1(b)(iv)	$a = r\omega^2$ **and** $\omega = 2\pi / T$	C1	**Allow** $a = v^2 / r$ **and** $v = 2\pi r / T$
	$T = 2\pi \sqrt{[(2.11 \times 10^{-10}) / (5.68 \times 10^{21})]}$ $= 1.21 \times 10^{-15}$ s	A1	Possible ECF from **1(b)(iii)**. Correct to at least 3 significant figures. AFC. All permutation of unrounded values give the same 3 SF answer.

<table>
<tr><th>Question</th><th colspan="2">Answer</th><th>Marks</th><th>Guidance</th></tr>
<tr><td rowspan="3">1(c)</td><td>radius:</td><td>decreases</td><td>B1</td><td rowspan="3">
<table>
<tr><th></th><th>decreases</th><th>same</th><th>increases</th></tr>
<tr><td>r</td><td>✓</td><td></td><td></td></tr>
<tr><td>a</td><td></td><td></td><td>✓</td></tr>
<tr><td>ω</td><td></td><td></td><td>✓</td></tr>
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</td></tr>
<tr><td>acceleration:</td><td>increases</td><td>B1</td></tr>
<tr><td>angular speed :</td><td>increases</td><td>B1</td></tr>
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